Evaluating RAG with HotpotQA

A step-by-step lab with LangGraph, sentence evidence, and scores you can inspect.

Find the evidence. Answer the question. Measure each step.

Basic Python and JSON · 60–90 minutes

Offline teaching path + optional live language-model experiment

Open the companion notebook while following the lesson.

Retrieval augmented generation

RAG gives an answer generator external text to use as evidence.



Evidence available

Evidence selected

Evaluation checks both the information found and the answer produced.

A **retriever** selects passages. A **generator** answers using those passages.

A plausible answer alone cannot tell us whether retrieval worked.

LangGraph agentic RAG

What HotpotQA contains

A 2018 question-answering dataset built from Wikipedia with sentence-level evidence labels.

113k

Question-answer pairs in the original dataset, approximately.

Questions often require combining information from two supporting paragraphs.

One labeled example includes

- A question and short reference answer.
- Candidate context paragraphs.
- Supporting facts: the sentence coordinates needed for the answer.
- Question type and difficulty metadata.

HotpotQA provides both answer targets and evidence targets for evaluating RAG.

Yang et al., 2018 HotpotQA project

Bridge questions

One fact identifies an entity that helps locate the next fact.

Question: In which city was the creator of Lantern Atlas born?



Bridge entity

Each hop supplies a fact needed by the next part of the question.

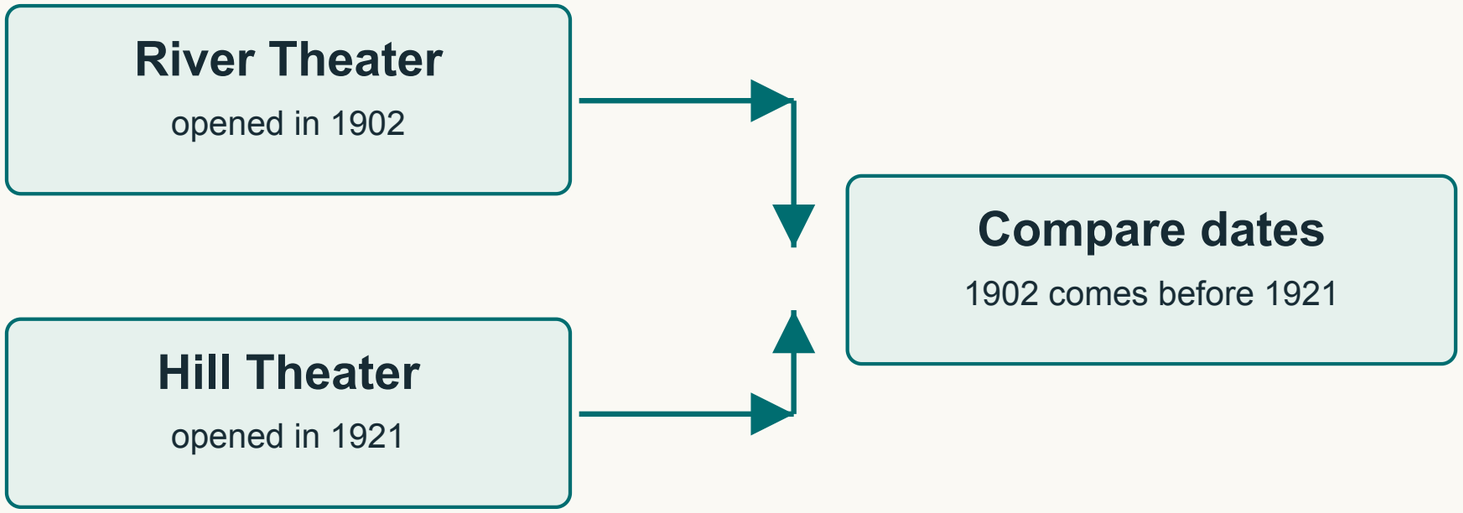
Invented teaching example. These names and facts are synthetic.

The first paragraph supplies the bridge entity “Mira Vale.”

Comparison questions

Two sources provide values that the system must compare.

Question: Which theater opened first, River Theater or Hill Theater?



Invented dates and theaters, used only to explain comparison.

Both relevant entities may already appear in the question.

Yang et al., 2018 Official JSON format

Distractor and fullwiki settings

The candidate collection changes what the experiment measures.

Distractor



10 supplied paragraphs

2 gold + 8 distractors

Fullwiki

Wikipedia corpus

Retrieve your own candidates

Gold paragraphs can be missed

Top-10 JSON ≠ whole corpus

This lab ranks each example’s supplied distractor paragraphs. It does not build a full Wikipedia index.

Results depend on the search space. Name that search space in every report.

HotpotQA project Official JSON format

A complete, small raw JSON record

The same synthetic record appears in the notebook and in the linked raw file.


```
{
  "_id": "synthetic-lantern-atlas-001",
  "question": "In which city was the creator of Lantern Atlas born?",
  "answer": "Harbor City", "type": "bridge", "level": "toy",
  "supporting_facts": [["Lantern Atlas", 0], ["Mira Vale", 0]],
  "context": [
    ["Lantern Atlas", ["Lantern Atlas is an invented book whose creator is Mira Vale."]],
    ["Mira Vale", ["Mira Vale was born in Harbor City."]],
    ["Harbor City", ["Harbor City is an invented coastal city."]],
    ["Cloud Almanac", ["Cloud Almanac is an invented book created by Niko Fern."]]
  ]
}
```


The real dataset file is a JSON list of records with this structure.

Official JSON format

The record has two kinds of information

The agent needs the task and candidate evidence. The evaluator needs the answers.

Field	Meaning	Use in this lab
`question`	The question to answer	Agent input
`context`	Title and sentence list per paragraph	Agent input
`answer`	The reference short answer	Scoring only
`supporting_facts`	Reference sentence coordinates	Scoring only
`_id, type, level`	Identity and descriptive metadata	Join results and analyze slices

Do not give the agent the entire labeled record.

Official JSON format

A supporting fact is an address

The pair [title, sentence_id] locates one original sentence.

Paragraph title: Mira Vale

0	Mira Vale was born in Harbor City.
1	A later sentence, if present, keeps index 1.

Citation coordinate: ["Mira Vale", 0]

`["Mira Vale", 0]` selects the first sentence, because indexing starts at zero.

Keep original titles and indices stable through ranking, chunking, and generation.

Official JSON format Official evaluator

An actual HotpotQA record

Selected fields from data/example.json, with original strings and coordinates.


```
{
  "_id": "5abd94525542992ac4f382d2",
  "question": "2014 S/S is the debut album of a South Korean boy group that was formed by who?",
  "answer": "YG Entertainment",
  "type": "bridge",
  "supporting_facts": [["2014 S/S", 0], ["Winner (band)", 0]]
}
```


["2014 S/S", 0]: 2014 S/S is the debut album of South Korean group WINNER.

["Winner (band)", 0]: Winner (Hangul: 위너), often stylized as WINNER, is a South Korean boy group formed in 2013 by YG Entertainment and debuted in 2014.

The full raw record retains all ten context paragraphs and every sentence.

Official JSON format Yang et al., 2018

The real sample uses the same schema

The notebook loads an attributed development sample and inspects its original records.

```
examples = json.loads(Path("data/dev_sample.json").read_text())
example = examples[0]
print(example["_id"], example["type"])
print(example["question"])
print(example["supporting_facts"])

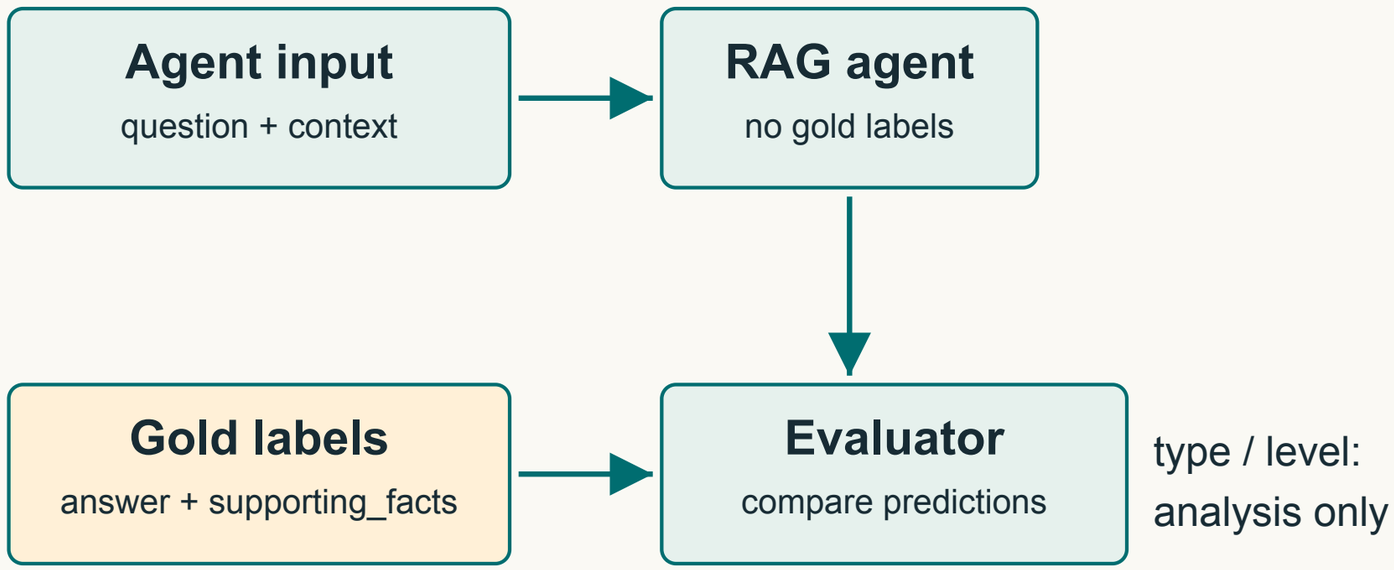
for title, sentences in example["context"]:
    for sentence_id, sentence in enumerate(sentences):
        print(title, sentence_id, sentence)
```

Inspect a complete record before changing preprocessing or retrieval.

Official JSON format

The agent cannot see the answer key

Gold labels enter the evaluator after prediction finishes.



The graph state and model prompts contain question, retrieved text, and run metadata.

A concrete experiment protocol

The default experiment evaluates retrieval within each real example’s distractor context.

1. Freeze the included example IDs and source checksum.
2. Build one paragraph index for each example.
3. Run the same question with `k = 1, 2, 4, 10`.
4. Score retrieval coverage against hidden gold labels.
5. Inspect failures before trying the optional live agent.

The bundled sample teaches the method. It cannot support a leaderboard claim.

Notebook setup

The required path runs locally after Python dependencies are installed.

```
cd hotpotqa
uv sync --locked
uv run jupyter lab

# Open the module notebook and run cells in order.
# Keep live model calls disabled for the offline lesson.
```

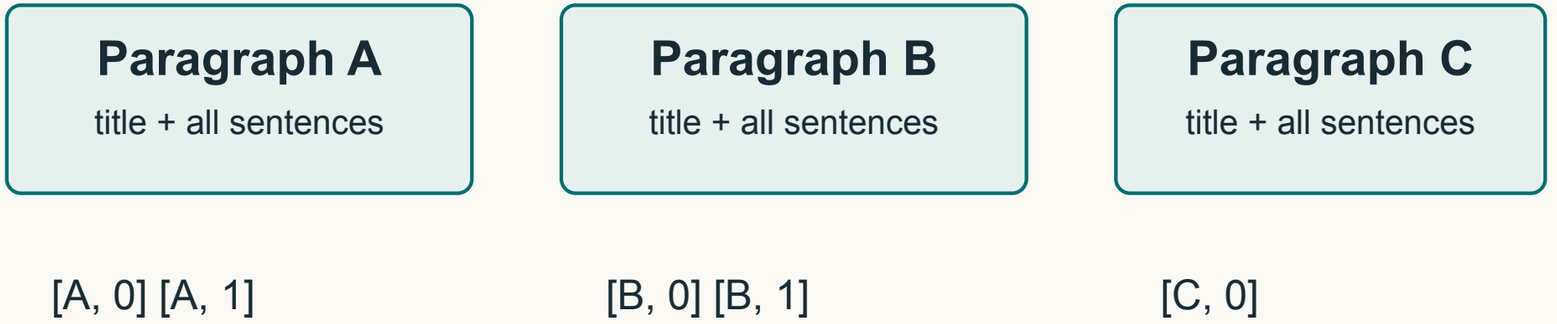
Use the module README for the notebook filename and optional model setup.

A fresh kernel avoids silently reusing old data, objects, or outputs.

Run the offline path first, then decide whether to enable live calls.

Paragraphs are the retrieval unit

Keep sentence coordinates inside each paragraph.



Rank paragraphs first. Ask the generator to cite individual sentences.

```
index = make_index(example["context"])
```

A paragraph can contain useful evidence and irrelevant sentences together.

A transparent TF-IDF baseline

TF-IDF represents a paragraph using weighted vocabulary terms.

Shared vocabulary: lantern, atlas, creator, mira, vale

Question	[0.5, 0.5, 0.7, 0.0, 0.0]
Paragraph A	[0.4, 0.4, 0.0, 0.5, 0.5]
Paragraph B	[0.0, 0.0, 0.0, 0.7, 0.7]

Words absent from the question can make the second hop hard to rank.

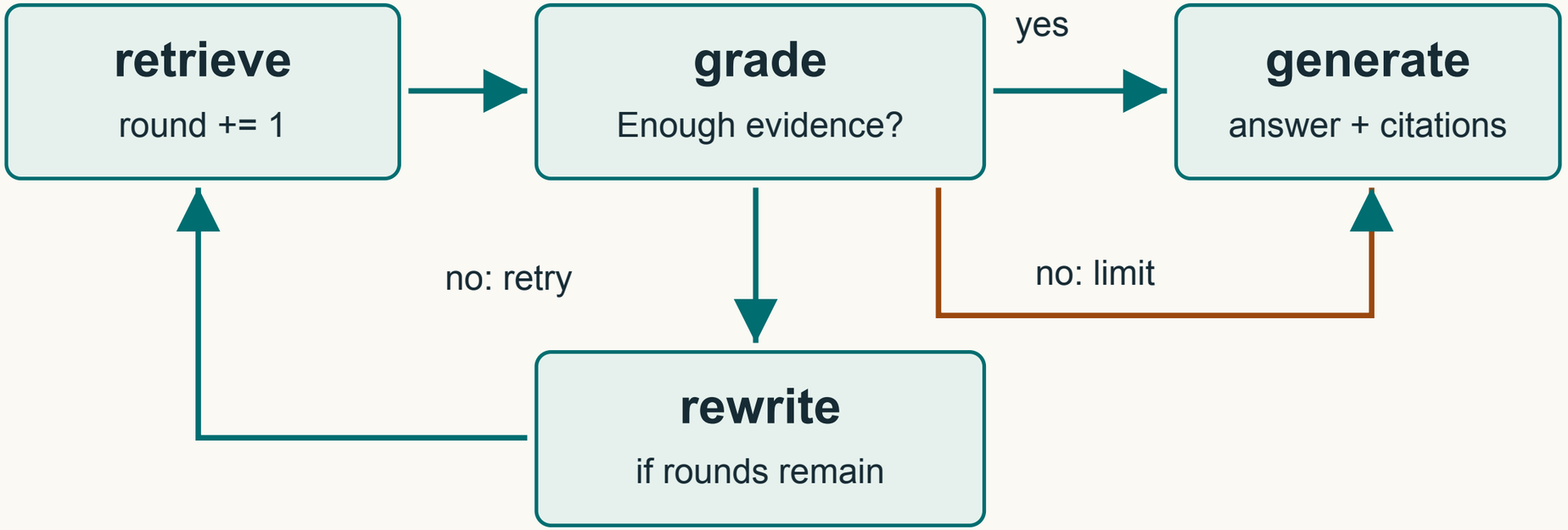
The displayed weights illustrate the idea. They are not values measured by the notebook.

This is sparse lexical retrieval. Dense embedding retrieval is a later experiment.

scikit-learn TF-IDF

The LangGraph workflow

Nodes read shared state, return updates, and route to the next step.



A fixed retrieval-round limit makes the loop finite.

LangGraph agentic RAG

What the graph remembers

State connects retrieval decisions to observable evaluation records.

Inputs and evolving evidence

- Original question and current query.
- Retrieved paragraph union.
- Round count and sufficiency decision.
- Execution trace.

Output for the evaluator

- Short `answer` and `sp`.
- `retrieved_titles` and `retrieved_facts`.
- `invalid_citations` and status.
- Latency and backend call count.

Save the trace so a score can be connected to the decisions that produced it.

Sufficiency is a routing decision

The grader checks whether the retrieved text can answer the whole question.

> **Question:** In which city was the creator of Lantern Atlas born?
>
> **Evidence:** “Lantern Atlas is an invented book whose creator is Mira Vale.”
>
> **Decision:** Insufficient. The creator is identified, but the birthplace is missing.

The grader receives the question and retrieved text, with no reference answer or supporting-fact labels.

A model grader can make mistakes. Its decision is not an official HotpotQA metric.

A rewrite can reveal the second hop

Use only facts already available to the agent when forming the next query.

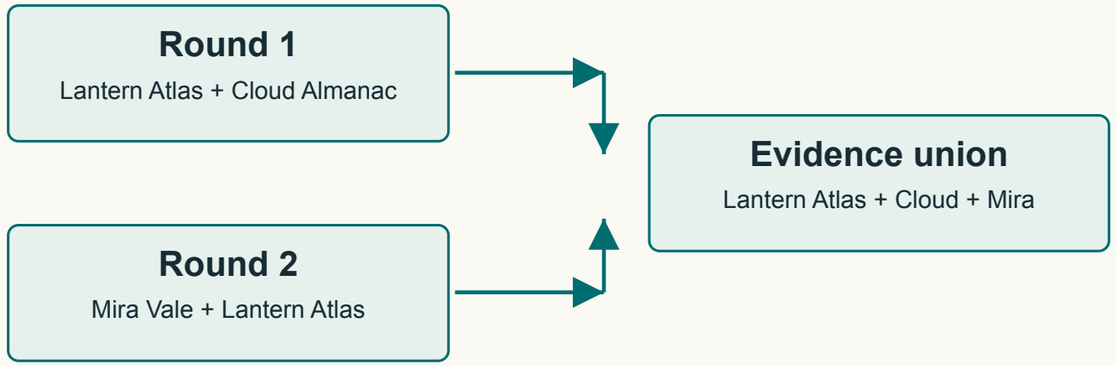
Initial query

In which city was the creator of Lantern Atlas born?

Rewritten query

Mira Vale birthplace

The name comes from retrieved text.



Two rounds with k = 2 can expose up to four distinct paragraphs.

The generator returns answer and citations

A short answer string keeps benchmark scoring separate from explanatory prose.

```
{
  "answer": "Harbor City",
  "citations": [
    {"title": "Lantern Atlas", "sentence_id": 0},
    {"title": "Mira Vale", "sentence_id": 0}
  ]
}
```

The adapter validates coordinates and converts citations into HotpotQA's `sp` pairs.

A citation must identify evidence the generator actually received.

Scripted demonstration and live experiment

The same graph can run with different backends.

	Offline scripted backend	Optional live backend
Decisions and answer	Provided by tutorial code	Generated by a model
What it demonstrates	Routing, retries, citation checks	Observed model behavior
API calls	None	Up to 4 per example at 2 rounds
Valid conclusion	The pipeline behaves as tested	Performance on this measured sample

A scripted correct answer is not evidence of RAG answer quality.

Three measurements, three meanings

Each metric answers a different question about one run.

Retrieval	Supporting facts	Answer
What reached the generator?	Which sentences were cited?	What did the system say?
Custom diagnostics	Official SP scores	Official answer scores

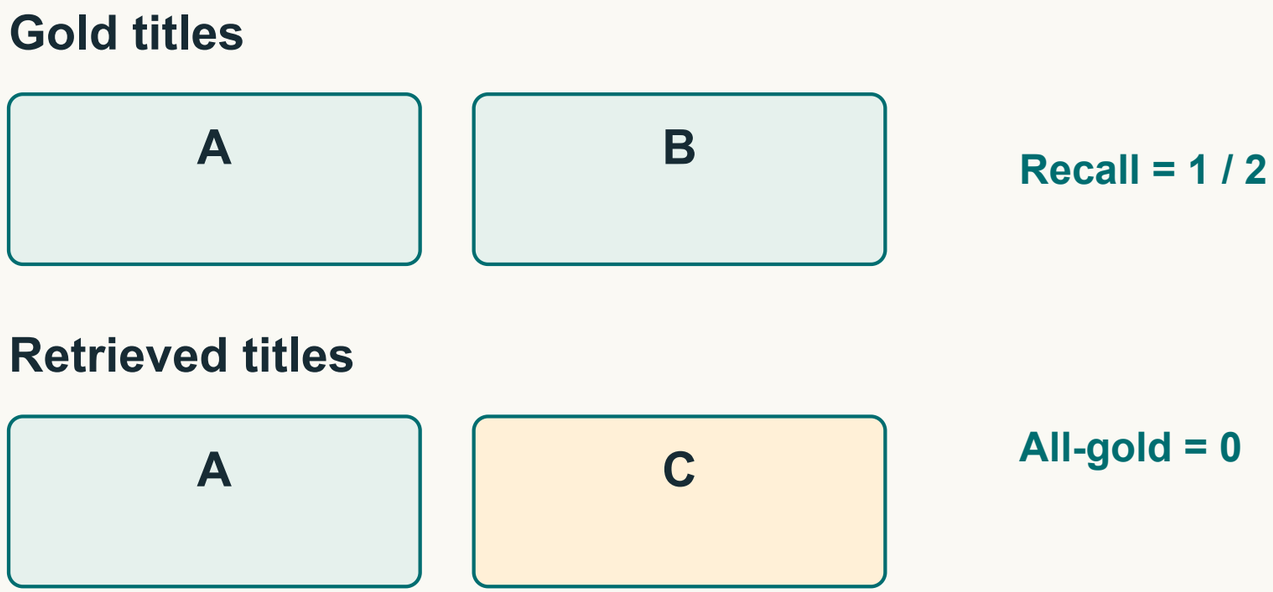
Joint scores combine answer and SP performance.

Official HotpotQA scores cover answers and supporting facts. Retrieval diagnostics are additional.

Official evaluator

Retrieval recall and complete coverage

Use the hidden gold facts only after retrieval to measure what was found.



Sentence coverage: gold sentence coordinates present in retrieved evidence ÷ all gold coordinates.

Finding one of two supporting paragraphs can still leave a question unanswerable from evidence.

Answer exact match and token F1

The official scorer normalizes text before comparing answers.

Gold: Harbor City

Prediction: Harbor

Precision = 1/1 = 1

Recall = 1/2 = 0.5

F1 = 2/3 ≈ 0.667

Exact match = 0

Normalization

Lowercase, remove ASCII punctuation, remove “a”, “an”, “the”, and collapse whitespace.

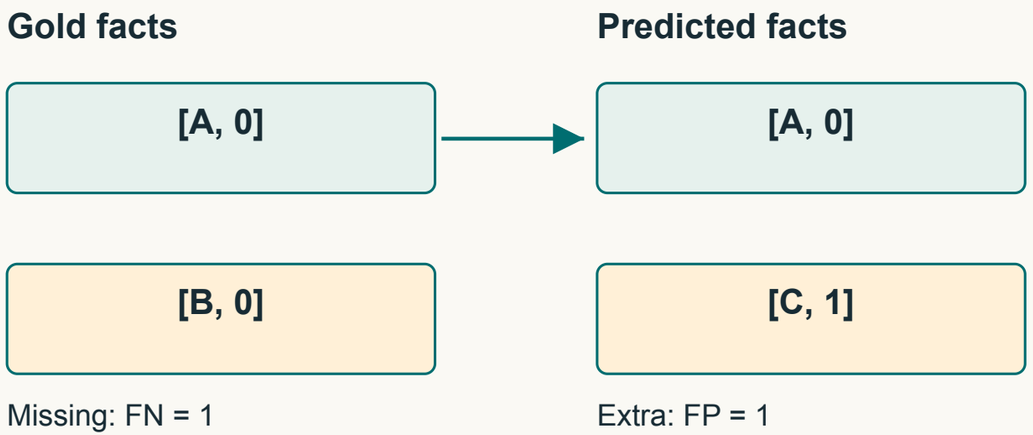
Token overlap counts repeated words. It does not measure semantic equivalence.

Special answers yes, no, and noanswer get zero token F1 when they mismatch.

Official evaluator

Supporting-fact precision and recall

Compare sets of exact (title, sentence_id) pairs.



TP = 1, FP = 1, FN = 1

Precision = 1/2

Recall = 1/2

SP F1 = 0.5

SP exact match = 0

Citing extra sentences can raise coverage while reducing evidence precision.

Official evaluator

Joint F1 combines precision and recall

Continue the previous worked examples: answer P = 1, R = 0.5; SP P = 0.5, R = 0.5.

Joint precision = answer P × SP P = 1 × 0.5 = 0.5

Joint recall = answer R × SP R = 0.5 × 0.5 = 0.25

Joint F1 = 2 × 0.5 × 0.25 / (0.5 + 0.25) = 1/3

Joint exact match = answer EM × SP EM = 0.

Compute per question, then take the mean. Use the formula above, even when a particular example happens to equal the product of the two F1 scores.

Joint F1 is generally not answer F1 multiplied by SP F1.

Official evaluator

The official prediction file

Two dictionaries use the original example ID as their key.

```
{
  "answer": {
    "synthetic-lantern-atlas-001": "Harbor City"
  },
  "sp": {
    "synthetic-lantern-atlas-001": [
      ["Lantern Atlas", 0], ["Mira Vale", 0]
    ]
  }
}
```

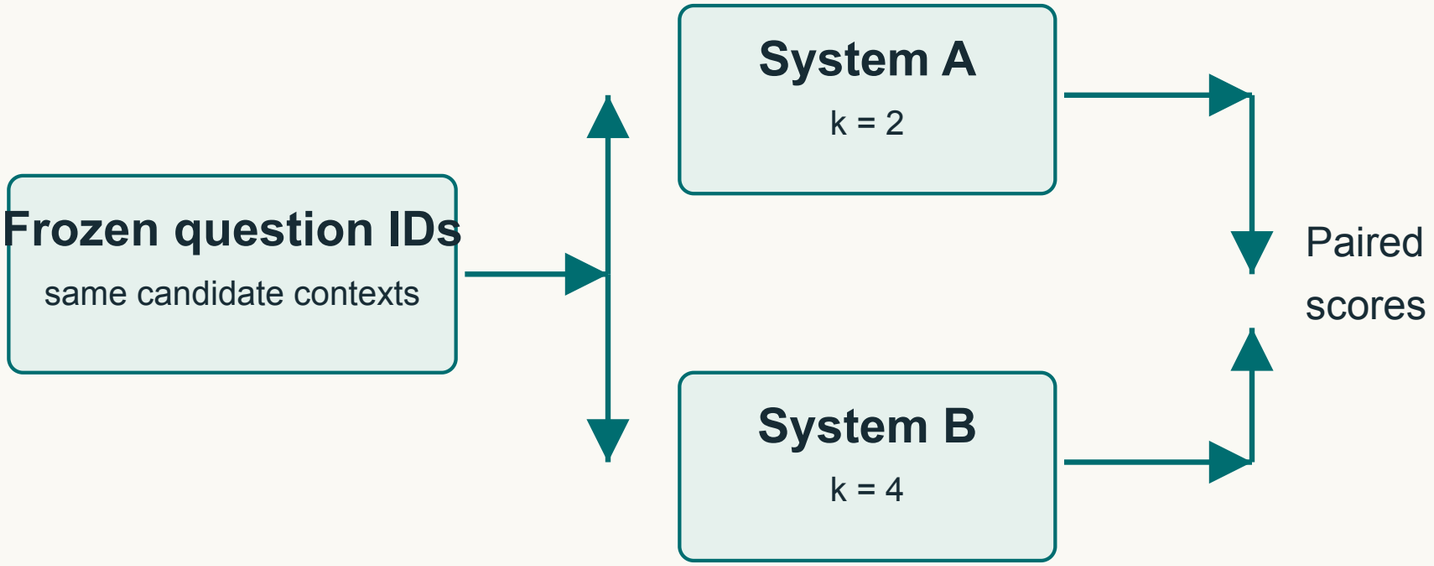
Synthetic format example. Real exports use the selected real HotpotQA IDs.

Score against the matching gold subset and keep its IDs with the predictions.

Official evaluator Official JSON format

A fair configuration comparison

Change one setting and retain per-question results.

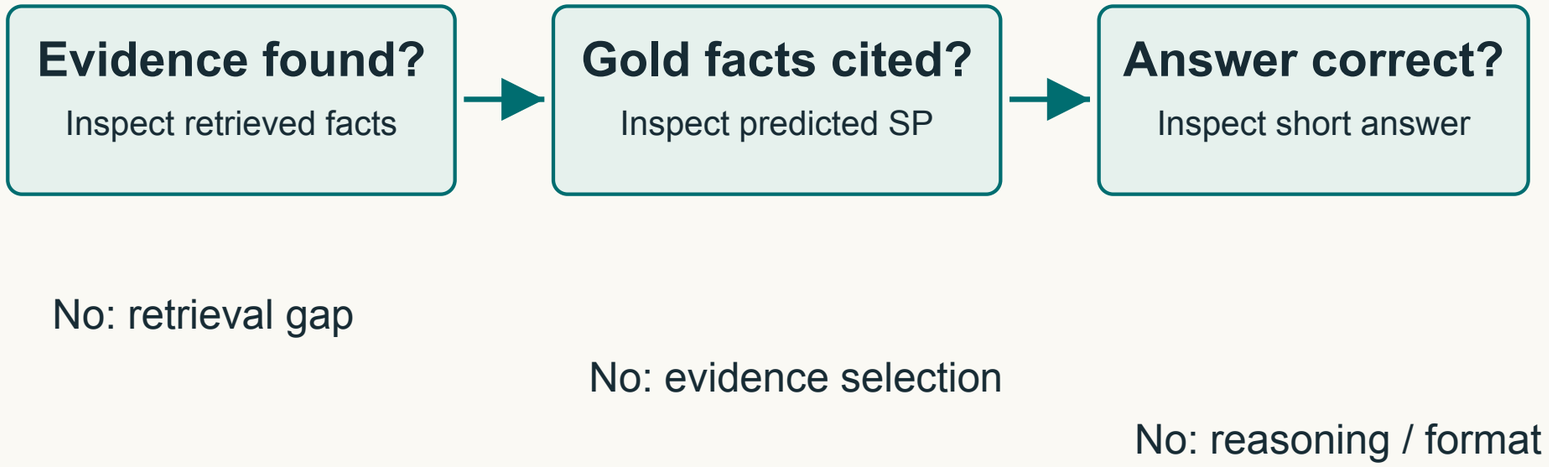


Report retrieval coverage, answer and SP scores where applicable, failures, latency, and model-call cost.

More retrieved paragraphs can improve coverage and also add distracting text.

Failure analysis follows the evidence

Use the saved trace to locate where the run first lost the needed information.



Also inspect invalid citations, malformed outputs, API errors, and runs that reached the round limit.

A final score summarizes a failure. The trace helps explain it.

A run needs an experiment record

Save enough information to reproduce the input and interpret the output.

Record	Why it matters
Source URL, checksum, IDs, split	Identifies exactly which data was scored
Candidate scope, k, round bound	Defines the retrieval task and evidence budget
Package lock, code, prompts, model	Identifies the implementation and model settings
Predictions, traces, errors, timing	Makes every scored outcome inspectable
Scorer version and metric definitions	Prevents silent changes in score meaning

A seed does not make hosted model outputs perfectly reproducible.

What a HotpotQA result does not establish

Benchmark accuracy is one part of an application evaluation.

- Wikipedia questions do not represent every production domain.
- Public benchmark material may appear in a model’s training data.
- Exact-string metrics can penalize valid aliases or alternative evidence.
- Supporting-fact agreement does not prove faithful model reasoning.
- A small distractor sample does not measure fullwiki retrieval.

Combine benchmark scores with application-specific tasks and evidence checks.

Yang et al., 2018 Official evaluator

Three short experiments

Predict the outcome before running each notebook exercise.

1. **Coverage:** Change k from 1 to 2. Which questions gain complete gold-paragraph coverage?
2. **Citations:** Keep “Harbor City” as the answer but remove one citation. Which official scores change?
3. **Loop:** Compare a scripted positive grade with a negative-then-positive sequence. Which nodes run, and how many paragraphs can accumulate?

Explain each change using saved evidence, not only the final mean.

A complete evaluation report

State the task, show the measurements, and explain their limits.

Task: Per-example HotpotQA distractor retrieval on a documented development subset.

Method: TF-IDF paragraphs, bounded LangGraph retrieval loop, backend clearly identified.

Evidence: Frozen IDs, prediction file, traces, coverage diagnostics, and official scores for genuine model outputs.

Interpretation: Failure categories, uncertainty, cost, and the next controlled experiment.

The notebook turns this report outline into executable steps.

HotpotQA project Yang et al., 2018 Official JSON format Official evaluator LangGraph agentic RAG